

Lab #4: DC-DC boost

Name: Tony Han

Table partners: Pranay, Theo, Marnu

#1. Experiment description and Prelab

##1.1 Description

In this lab, I will explore a dc-dc boost circuit which we discussed in the tutorial prior. The objectives are to understand how to use more circuit components allowing for me to build more advanced circuits, utilize energy storage in inductors to boost voltages, and understand how loads affect circuit behaviour.

In this lab, deliverables include actually building the dc-dc boost circuit and operating it with some given initial parameters in the lab instruction document. Then, I will perform a series of measurements:

1. Take time-dependent voltage measurements across the various components of the circuit, including MOSFET input, then plot it on the same graph to confirm operation of the circuit. Then use this to explain and describe what the circuit does, and how it does it. Note differences between my prelab ideas and reality.
2. Alter the frequency (in a range of $5\text{kHz} - 25\text{kHz}$) of the square wave driving the MOSFET and observe changes to V_{out} . Note differences between the 30kHz case, and explain with my understanding of how the circuit works, what is going on.
3. Measure/calculate the average input current from V_{in} and the average output current across R_L and then calculate the power efficiency of my circuit. (Take $R_S = 0.1\Omega$ as given)
4. Change R_L to 50Ω , observe and explain changes.

##1.2 Prelab

Here, I will explain how a DC-DC boost circuit such as the one provided to us, works. Below, I have attached the circuit diagram that was provided in the lab instruction document.

In [3]: `InlineImage('dcboost.png', width=12, height=12, caption= "Figure 1, prelab circuit diagram of a DC-DC boost cir`

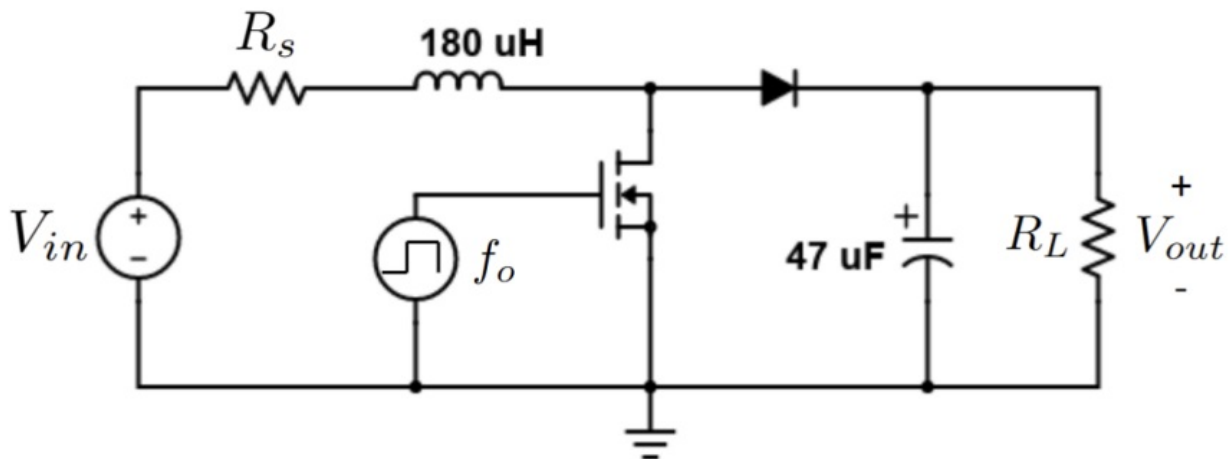


Figure 1: DC-DC boost Circuit

Figure 1, prelab circuit diagram of a DC-DC boost circuit

First, I will describe the circuit: There is a constant voltage source, as well as a wave generator source attached to a MOSFET. Prior to the MOSFET, there is a resistor and an inductor. When the MOSFET is on, a short is caused in the circuit, so current skips the branch with the diode and goes to ground. When the MOSFET is off, the circuit passes into the branch with the diode and what is essentially a RC circuit. This resistor represents the load, and is what will take the higher output voltage from the DC-DC boost and put it to use. Note that the current can only enter this RC loop and not exit due to the diode. Also note that the capacitor has a polarity bias.

How does this circuit actually work? well, first, lets assume the square wave starts at a non-zero voltage that meets the threshold for the MOSFET to turn on. The circuit has a resistance R_S , which, along with V_{in} , sets the current. The inductor initially resists the current but eventually reaches steady state. In this time, I assume that the inductor stores its maximum possible quantity of energy.

Now, lets assume the square wave now changes so that the voltage is 0. The MOSFET turns off, and the current must flow to the branch with the diode, and the capacitor and load. Since the load increases the total resistance in the circuit, the current will drop, which the inductor wants to fight! The inductor now essentially acts as a battery to maintain the current as it was before the MOSFET was turned

off. The total voltage all gets added up and given to the capacitor, which then holds a higher potential than V_{in} due to the inductor's contribution. And since there is a diode at the entrance of the RC loop, current (charge) does not exit; it stays in the loop. The capacitor holds all of this charge, and given that $Q = CV$: if Q increases for a given C , so must V . Hence the DC-DC boost. The inductor stores energy, and acts as a second battery to charge the capacitor to higher and higher voltages. The capacitor supplies this higher voltage to the load, which is in parallel with the capacitor (so it must have the same voltage drop). Of course, some energy will be dissipated, so the voltage may drop a bit. But, not to worry; this is when the square wave changes yet again so that the MOSFET turns on, and the whole process repeats. The inductor is charged again, and then when the MOSFET closes, it gives its energy to the capacitor on top of what V_{in} gives. And by parallel circuit behaviour, V_{out} is thus much higher than V_{in} ; so the load has access to a much higher voltage.

I may add an image of the different steps to aid this explanation post lab.

Below is a graph of what I expect the plot of voltage across the diode as a function of time in ideal circumstance, with a square wave input to the MOSFET with some period T_o :

In [5]: `InlineImage('dcgraphpng.png', width=12, height=12, caption= "Figure 2, prediction of voltage accross diode as a`

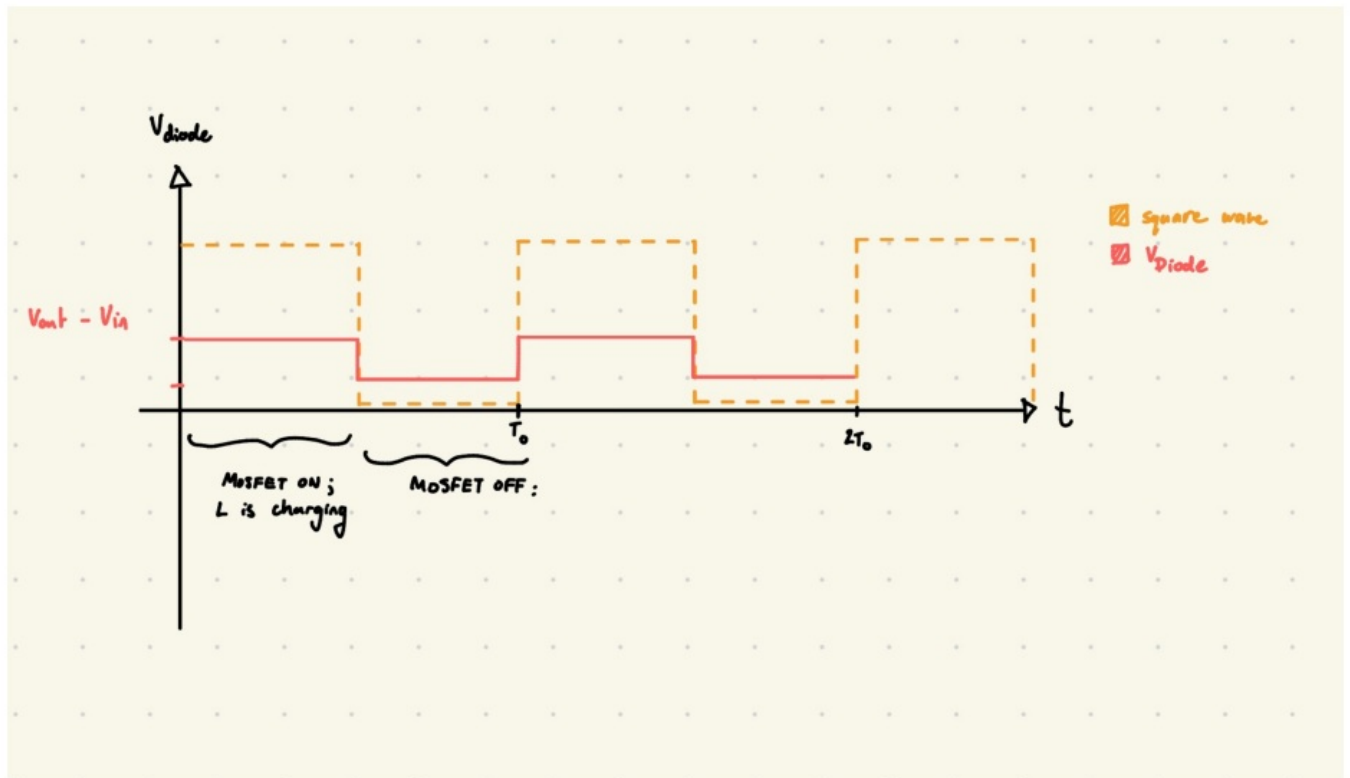


Figure 2, prediction of voltage accross diode as a function of time for ideal DC-DC boost circuit

#2. Experiment

##2.1 Circuit Construction

Here, I will build the circuit. Below is my circuit diagram and finished circuit. I measured the shunt resistor to be 0.092Ω . However, as the TAs mentioned, this is more so for curiosity rather than numerical value as we don't really have uncertainty to propagate in this lab.

In [7]: `InlineImage('loadsetup.png', width=12, height=12, caption= "Figure 3, Initial setup of circuit, for DC-DC boost
InlineImage('bestCircuit.png', width=12, height=12, caption= "Figure 4, Circuit diagram for DC-DC boost circuit`

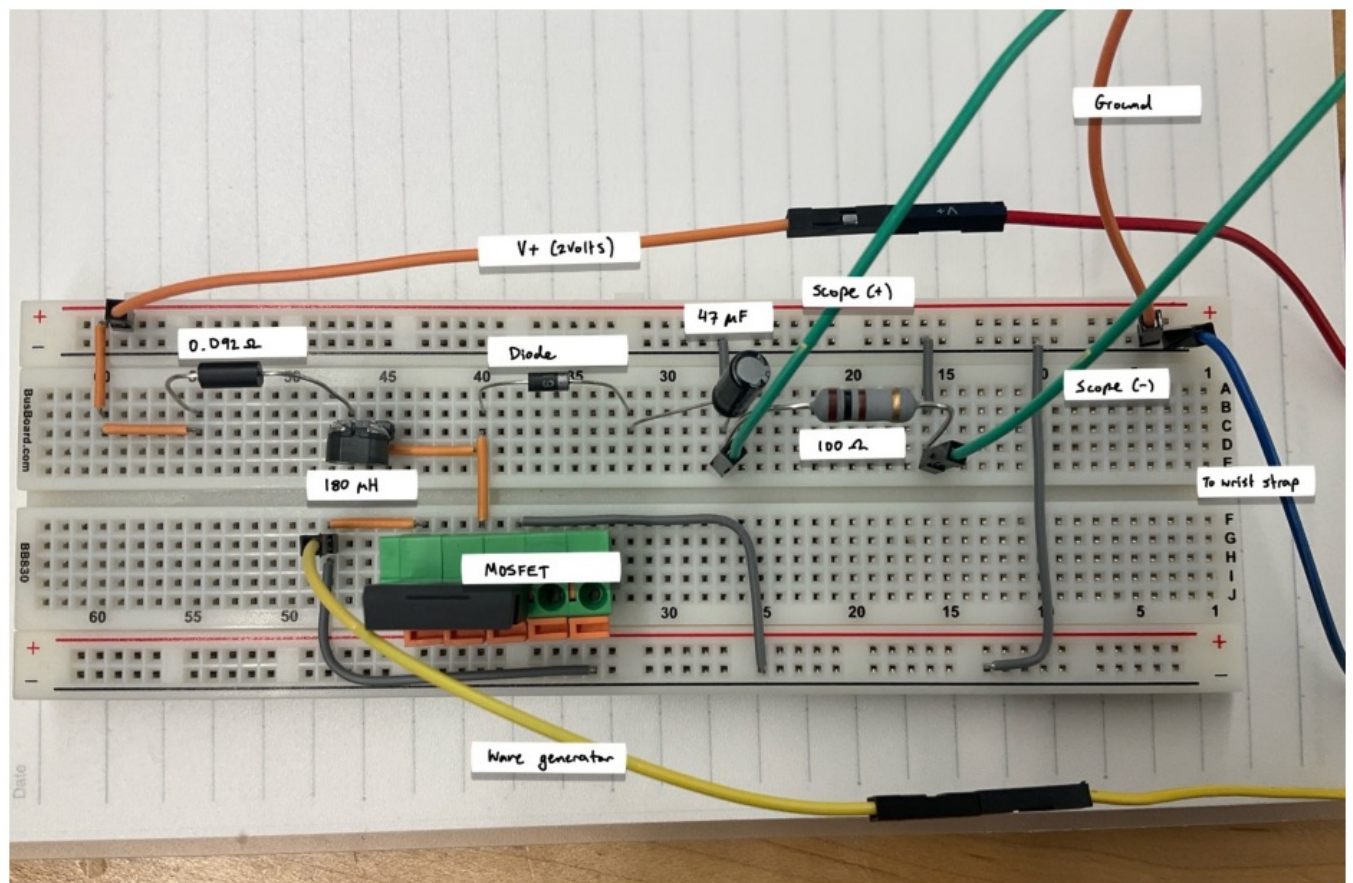


Figure 3, Initial setup of circuit, for DC-DC boost circuit.

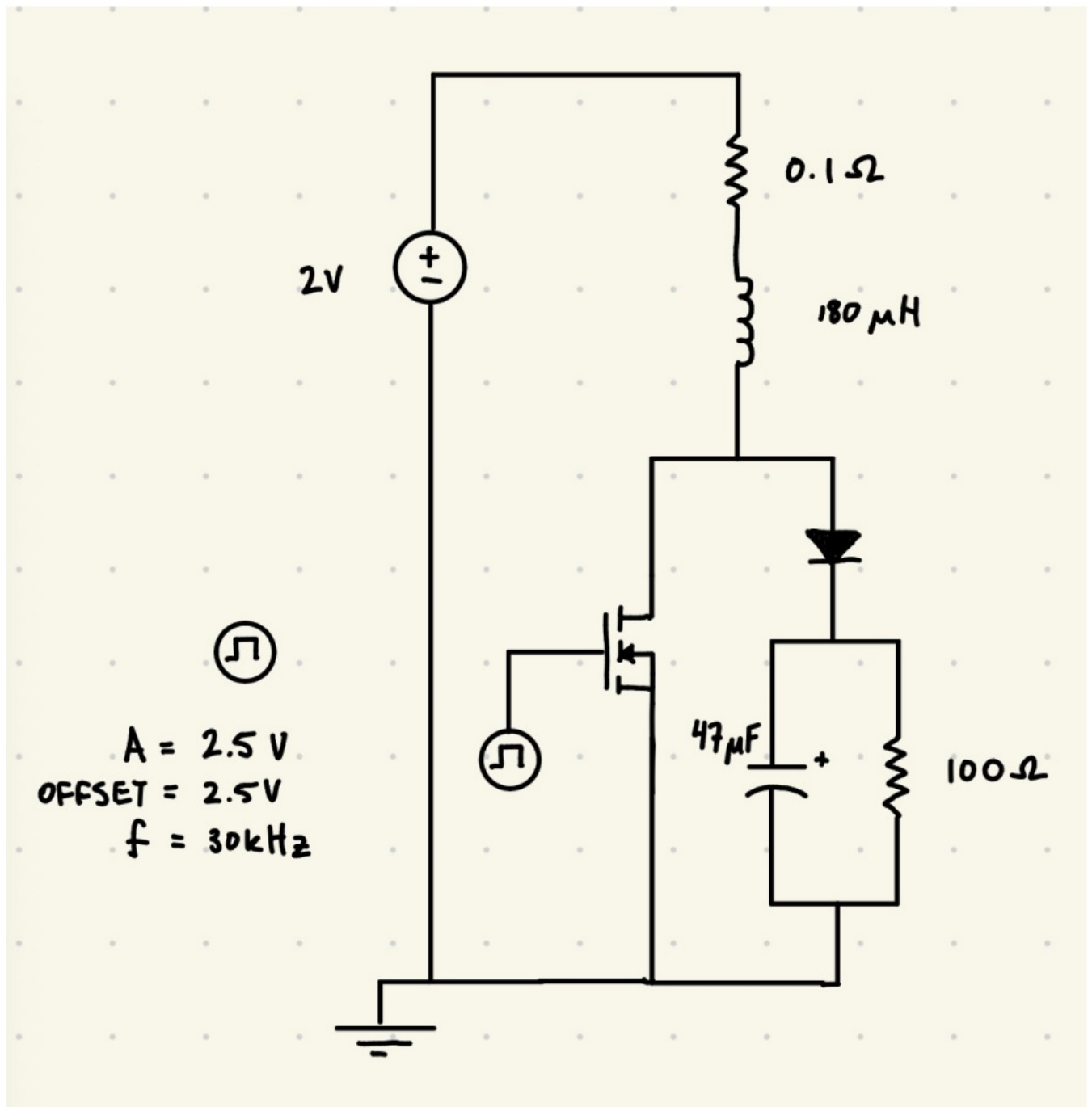


Figure 4, Circuit diagram for DC-DC boost circuit.

Strange, probing over the load resistor gives me no values. This is not supposed to happen

TS First, I will just verify that my circuit has been built according to the diagram in the prelab document. Whoops, turns out my load resistor and capacitor were in series rather than in parallel; this will cause the current to short to ground before reaching the load resistor. I have changed it, and now I get a reading on the scope.

TS The voltage over the load resistor reads as just under 2V, and 2V is my input; this means that my circuit is doing the opposite of a boost; it is decreasing the output voltage!! I am not quite sure what is happening. I have verified that my MOSFET works, and my circuit is built right. Maybe there is something I missed?

TS I used the probe of the second scope channel to see what is happening over the gate. I am getting the right voltage drop, so I know that my MOSFET is receiving the right input. So the switch portion should work. Everything else I also double checked to have the right connection. Diode is correct.

TS I asked the TA and we took another look at the circuit, turns out in arranging the circuit, the connection from my voltage source to the gate pin on the MOSFET adapter was on the wrong row. Fixing this provided me with the proper wave form. Below is a screenshot of this waveform, along with my new correct and working circuit.

In [11]: InlineImage('newsetup.png', width=12, height=12, caption= "Figure 5, Initial setup of circuit, for DC-DC boost")

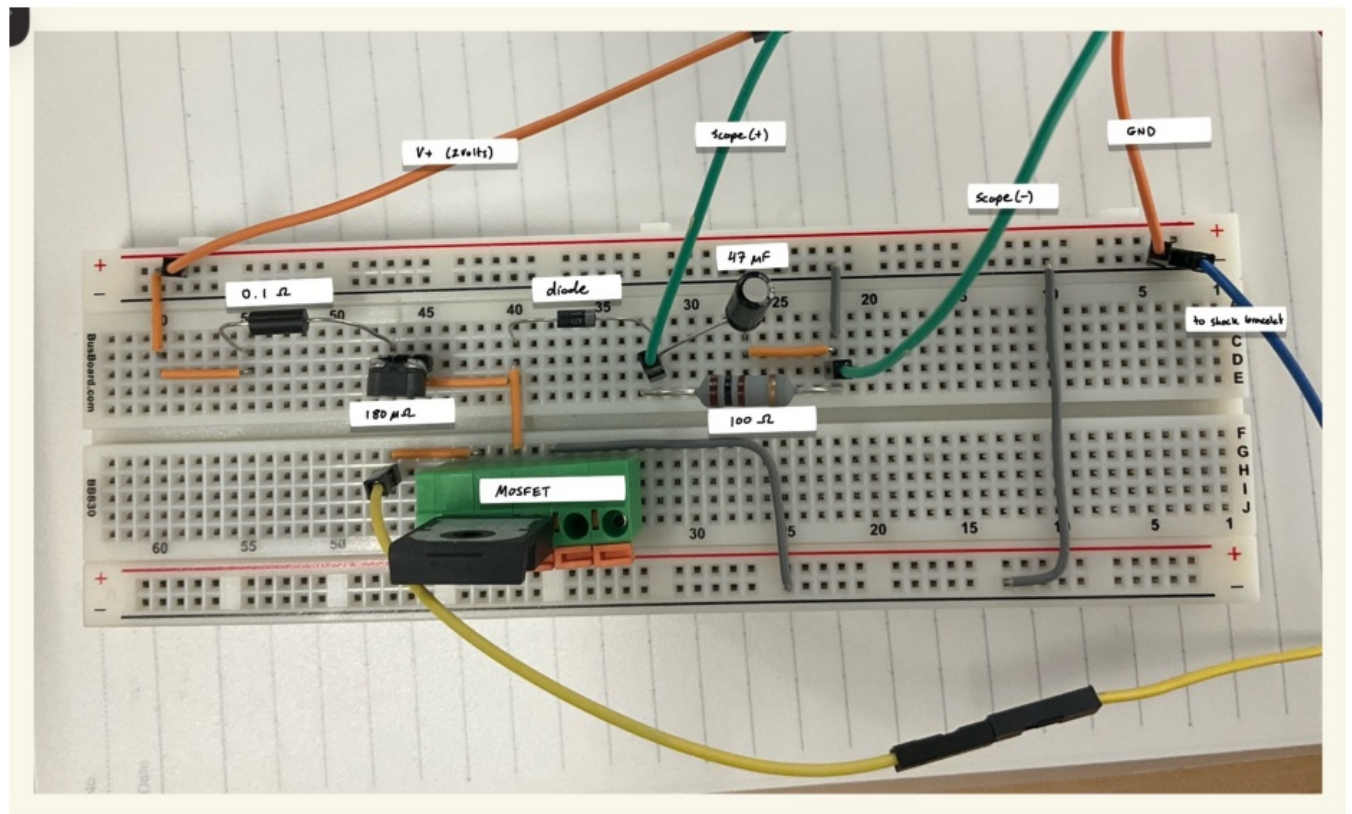


Figure 5, Initial setup of circuit, for DC-DC boost circuit.

Here are my source and wavegen settings:

```
In [16]: InlineImage('sourcesettings.png', width=12, height=12, caption= "Figure 6, Source settings", fs=12);
         InlineImage('wavesettings.png', width=12, height=12, caption= "Figure 7, wavegen settings", fs=12);
```

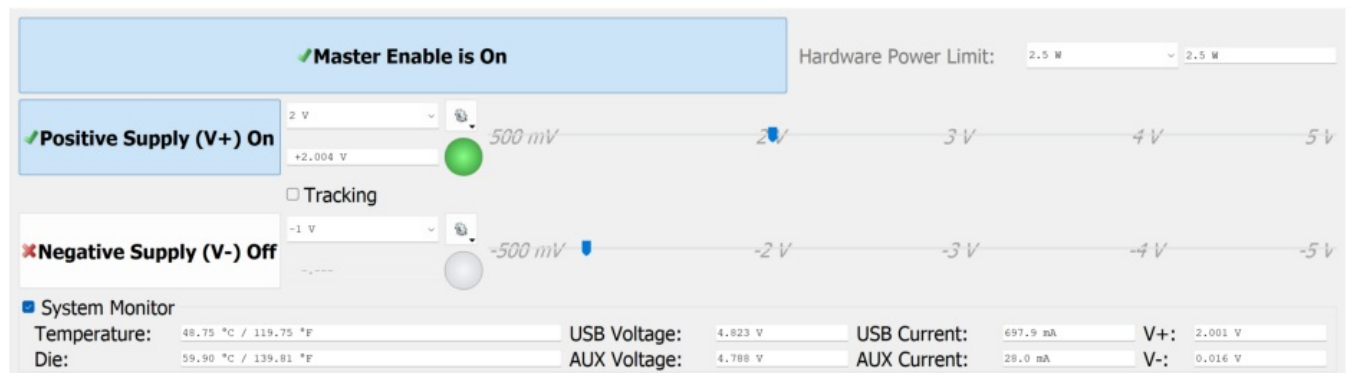


Figure 6, Source settings

Type:	□ Square	⚙
Frequency:	30 kHz	▼
Period:	33.333333333 us	▼
Amplitude:	2.5 V	▼
Offset:	2.5 V	▼
Symmetry:	50 %	▼
Phase:	0 °	▼

Figure 7, wavegen settings

Here is my successful trace of the voltage over the load resistor:

```
In [15]: InlineImage('tracedd.png', width=12, height=12, caption= "Figure 8, scope trace of voltage over the load resist")
```

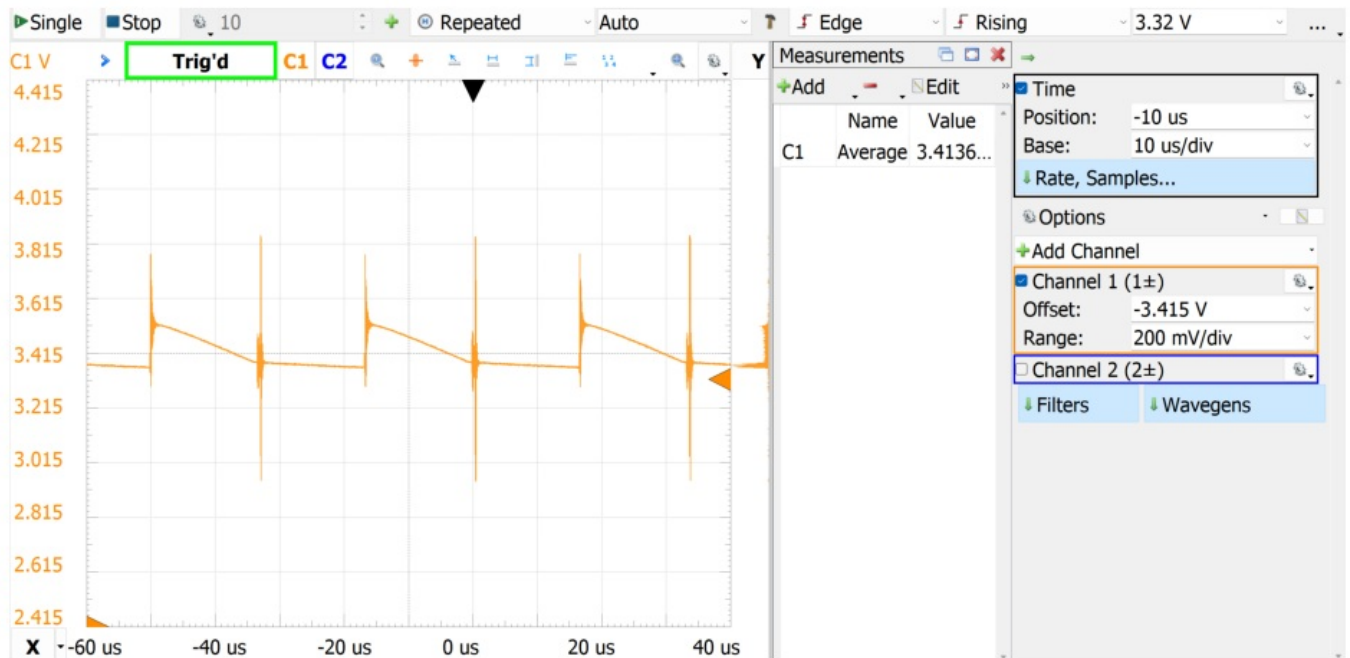


Figure 8, scope trace of voltage over the load resistor in a dc-dc boost circuit. Input is 2V, average output is 3.4V. Load resistance 100 Ω

Using the average, I measured the output voltage to be around 3.4 Volts, and this value fluctuates between a min/max based on whether the MOSFET is on or not. When the MOSFET is off, the voltage is at around 3.67 volts, and it decreases with a relatively linear rate. This is likely explained by the inductor providing extra current, but the amount of current it provides decreases as its stored energy gets drained. However, I think that the inductor drainage should be a downwards exponential (recall phys 158)..? so the voltage over the resistor should also follow that pattern??

I think I know why, after a lot of thinking:

I think that maybe it is incorrect to refer to the branch as simply an RLC circuit; it is an RLC branch connected to a constant 2V power supply. Because of this, the slope functions a bit differently. I will first explain what the waveform for current through the inductor should look like.

I will explain my understanding in two parts, one for when the MOSFET is on, and one for when it is off. These will be based off of the relationship:

$$\frac{di}{dt} = \frac{V_L}{L}$$

1. When the MOSFET is on, we know that the circuit gets short circuited such that the current does not flow through the diode. In this case, there is no additional voltage drop other than over the inductor so $V_L = V_{in}$. Because V_{in} is a constant value, we have that $\frac{di}{dt}$ must also be a constant value. So, when the MOSFET is on, it should be a positive linear slope for current through the inductor.
2. when the MOSFET is off, we know that the circuit now has a voltage drop over the load/capacitor in parallel. As such, $V_L = V_{out} - V_{in}$, which is yet another constant value. However in this case, since it is a voltage booster, we have V_L is negative. So when the MOSFET is off and there is current flowing through the diode, the current through the inductor should follow a linear negative trend.

With these two conclusions, I think that the current accross the inductor should follow a triangle wave ish pattern. As such, when the inductor's current goes through the diode (when the MOSFET is off), we should observe the same linear behaviour with the voltage over the resistor (a downwards linear trend). So this makes sense! Also, by principle of superposition, the constant current supplied from the 2V DC source does not affect the slope; only the min/max value.

It seems that, however, there is a flat(ter) part. I think this is probably when the MOSFET is on, and the voltage over the resistor is the same as the capacitor (the capacitor gives power when the rest of the circuit is cut off due to the MOSFET short circuit). There is a little bit of drop, I would assume due to power dissipation or essentially the system not being ideal.

All in all, this measurement and the general shape of the waveform makes sense.

However, I will say that I am unsure as to why there are such sharp jumps during switching of the MOSFET. I hypothesize that it is something to do with my trigger or oscilloscope reading sudden changes in voltage. That, or maybe interference from other parts of the circuit (such as the capacitor)? Lots of wonky stuff when we deal with switches that are sudden.

That is all I will say for this section. Maybe I will do my own research later.

##2.2 Tasks

###2.1.1 Internal Circuit Voltage Measurements

Here, I will take measurements of the voltage over the diode, inductor, and load, as well as the MOSFET input. Then, I will plot these in the same graph. In order to maintain consistency, I will measure each voltage drop with respect to the MOSFET input, to make sure the timing lines up for each measurement.

In [5]: `InlineImage('voltages1.png', width=16, height=12, caption= "Figure 9, Scope trace of voltage over Diode (red),`

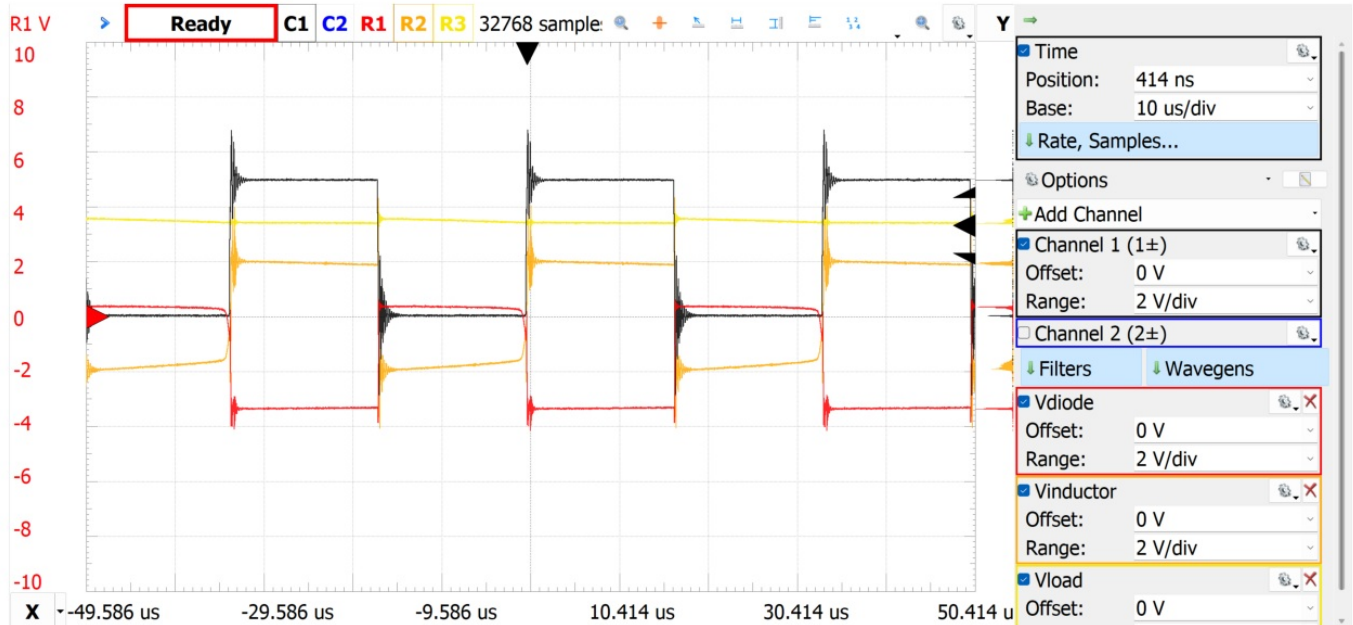


Figure 9, Scope trace of voltage over Diode (red), Inductor (orange) and load (yellow) taken in reference to the MOSFET input (black).

This graph is a lot to take in. First, I will confirm my theory above about the MOSFET stage and the linear ish decrease in voltage about the load resistor.

In yellow, we have the load resistor. As is visible, the sections with a visible and significant negative slope are when the MOSFET is off (aka MOSFET input voltage is 0). I think that this is good evidence that my suspicions about the linear decrease in voltage over the resistor is true. We can also see that the flatter part in the yellow graph for V_{load} occurs when the MOSFET is on, which also aligns with what I said earlier. SO that is one function of the circuit.

Next, I will talk about the voltage across the inductor.

Over the inductor, (in orange), the voltage fluctuates between -2V and +2V. There is a transition from -2V to +2V, but what seems like a sudden drop from +2V to -2V.

First, I will look at the -2V section. In this case, the MOSFET is off. During this period in the boost circuit, the current flows through the diode, and the inductor is discharging. At first, recalling inductor behavior from phys158, the inductor initially acts as a battery oriented accordingly to maintain current flow. In this case, given the load, the total resistance increases so the current wants to decrease. However, the inductor essentially acts temporarily as a battery (2v source in this case given the graph reading) to hold the original current. This imaginary battery has its positive terminal oriented towards the diode, which explains the negative polarity; I placed my nodes in the typical higher potential to lower potential, with the negative end on the side with the diode. Now, given that the inductor stores a finite quantity of energy, we expect a gradual decrease in voltage over time as the inductor discharges its stored energy. It will then be unable to maintain its "battery" and eventually deplete, functioning as a wire with no voltage drop.

On the other hand, for the drop from +2V section. In this case, the MOSFET is on. During this period in the boost circuit, current is short circuited through the MOSFET, and the inductor enters its charging phase. In this case, given that the other end of the inductor is connected to the ground, we have that the voltage drop should be equal to the source power. We also know that the shunt resistor has an extremely small resistance, meaning very small contribution to voltage drop; hence why in this case, the voltage looks like 2V.

Where this waveform differs from the ideal case is the gradual increase in slope when transitioning from -2V to 2V. This is what I think happens:

$V_L = V_{in} - V_{out}$, and V_{out} is decreasing (see the yellow trace for voltage over the load resistor). This means that V_L is decreasing over time. what does this affect? over time, as the voltage over the load resistor decreases (maybe lost power or current leak or unideal efficiency), V_L becomes less and less negative. This explains that positive slope over the longer portion. However, I am unsure of what causes that sudden exponential looking growth up to 2V. My guess is that the inductor runs low on stored energy or something like that and eventually works as a wire, no longer contributing as much to current flow...? (hence it going to 0V). Not sure.

Finally, will talk about the voltage across the diode.

Here, the voltage accross the diode is shown in red. The positive probe I placed towards the 2V DC source, the negative probe away. We see a slightly positive voltage drop (0.4V ish) when the MOSFET is off, and a larger negative voltage drop of approx. -3.6V when it is on. Note that, this negative voltage drop is the negative of the boosted voltage over the load resistor. There is a reason for this:

When the MOSFET is on, during the same time period the inductor has a voltage drop of 2V since it is connected to the 2V source and ground (KVL). The load has also been boosted to 3.6 ish volts. In order to satisfy KVL, we must have the voltage drop over the diode be -3.6V. This can work since the diode is not a traditional element where the voltage drop depends on some other variable such as current. Instead, it can have any value to satisfy KVL (just like current sources can have voltage values to satisfy KVL).

When the MOSFET is off, during the same time period, once again the diode voltage serves to satisfy KVL. The inductor has a voltage drop of -2 ish volts, while the load resistor has a drop of 3.6V ish still. Here, the diode has a drop of 0.4V, bringing the total to $(3.6+0.4)-2 = 2V$, which is the same as the power source. Hence, KVL is satisfied.

CORRECTION TO MY PRELAB PREDICTED WAVEFORM SINCE I WAS WRONG

For the diagram in my prelab, my predicted graph was slightly wrong. In particular, the polarity of the voltage during the MOSFET on stage. I got the voltages mixed up, and instead of satisfying KVL I violate it with my prediction. I think that this was partly due to it being late at night and me being tired/sloppy. However, the magnitude (ignoring signs) is similar to what I observed in the actual lab.

GENERAL OBSERVATIONS

When the voltages across all components at any given time add up to 2V, which is the same as the source!! this means that the circuit makes logical/physical sense. There is no violation of physics laws here.

###2.1.2 Drive Frequency

As I decrease the frequency of the square wave operating the MOSFET switch, I notice the following:

Frequency	min V	max V
25 kHz	3.18	3.82
20 kHz	3.29	4.05
15 kHz	3.44	4.34
10 kHz	3.56	4.79
5 kHz	3.39	3.81

At 5kHz, my ad3 started making a high pitched noise....

anyways, here is a summary of my actual results.

Decreasing the frequency more and more (more time between switch open/closed) increases the boosted voltage value... until I decrease to a frequency of 5kHz. This is strange behaviour, but I would expect that at a certain point, the voltage cannot be boosted any higher.

My hunch for the general trend (excluding 5kHz for now) is that, by decreasing the frequency, there is more time for the inductor to fully charge to the maximum quantity of energy it can possibly hold. Hence, the additional current can be sustained for longer and more energy can be passed to the capacitor, thus increasing the voltage.

However, the 5kHz case stumps me.

Observing the voltage over the diode shows me something: there is a sinusoidal waveform that takes place between the positive and negative portions of the original waveform. (starting from some point in MOSFET off and lasting until MOSFET on). This sinusoid portion initially does not take up a significant portion of the MOSFET off stage, but the portion it takes up increases significantly as the frequency gets to the lower values.

Below is an image of the sinusoid portion for voltage over the diode:

```
In [9]: InlineImage('sinudois.png', width=16, height=12, caption= "Figure 10. sinusoid waveform for voltage over the di")
```

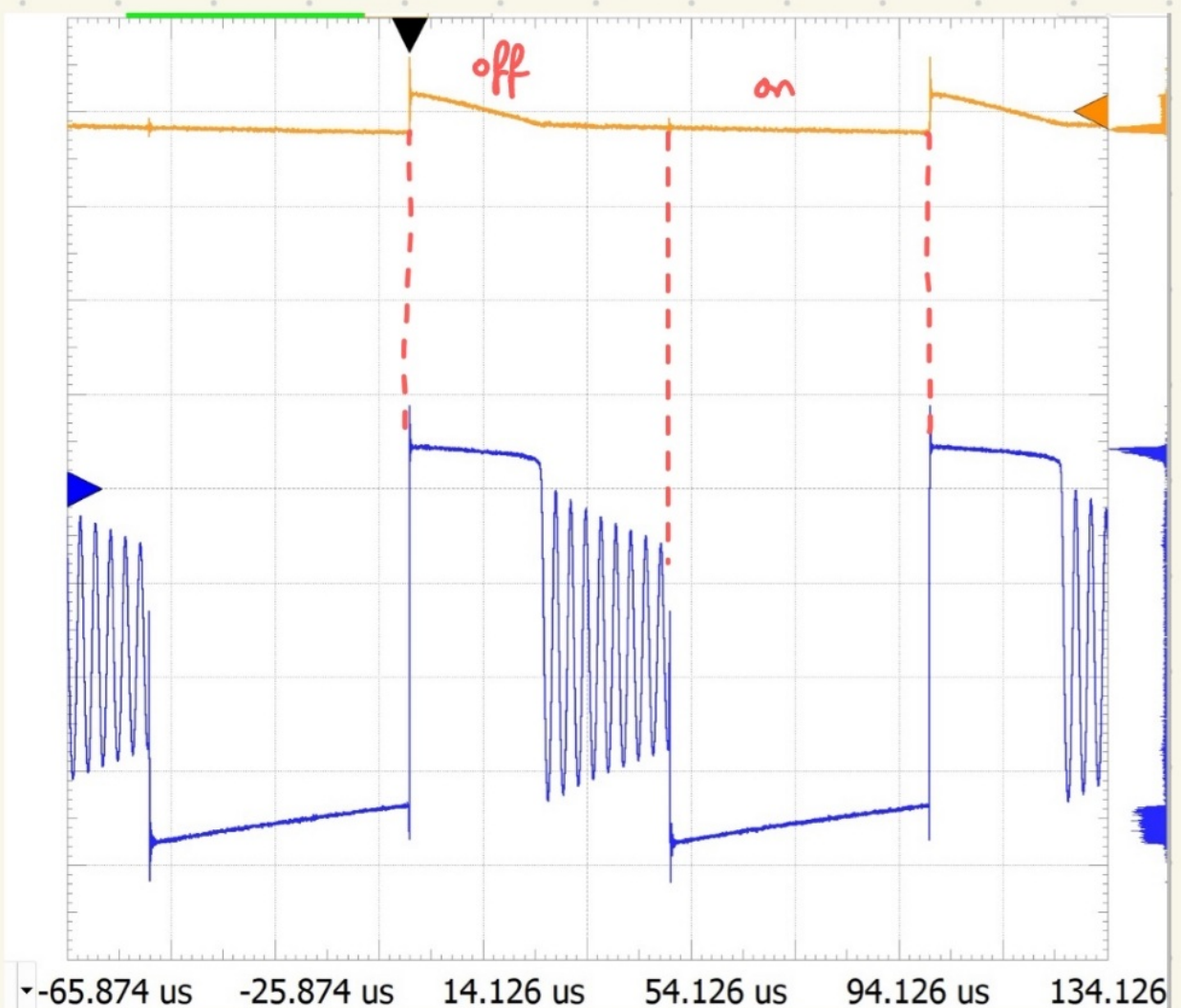


Figure 10. sinusoid waveform for voltage over the diode at lower frequencies. this example taken at 10kHz.

There is a similar result when I look at the voltage over the inductor; this sinusoidal waveform takes up more and more as I decrease the frequency.

Whatever causes this sinusoidal behaviour for voltage is likely the culprit; as this behaviour increases to a certain amount, the boost no longer boosts as high despite turning the frequency down.

Here is my hypothesis:

Increasing the period gives more time for the inductor to charge, but also to discharge. Since the sinusoidal portions occur when the inductor is discharging, I think that there may be something to do with the inductor being fully discharged and having no more energy left to provide.

Given that the voltage is fluctuating so much, I think that the current is also fluctuating in a similar way. (also note; once again, the sinusoidal portions of diode voltage and inductor voltage satisfy KVL).

As for the sinusoidal parts, this is probably where the inductor runs out of juice. I notice that in the case of the inductor, the fluctuation (and the entire waveform) is still symmetric about the x axis (ie. equal in amplitude on positive and negative):

In [13]: InlineImage('sinusod.png', width=12, height=12, caption= "Figure 11. sinusoid waveform for voltage across the i

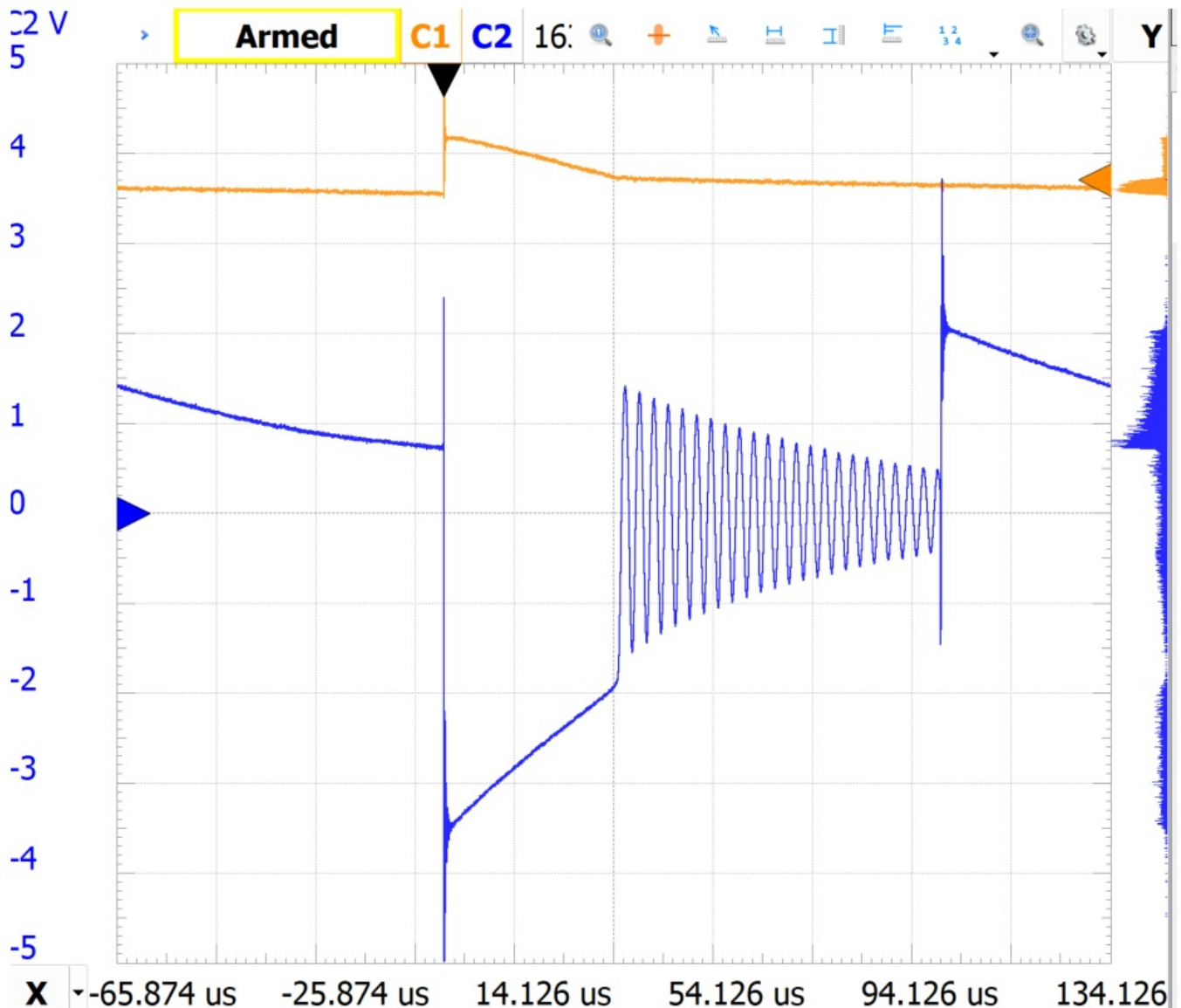


Figure 11. sinusoid waveform for voltage across the inductor (blue), 5kHz MOSFET input.

When V is positive, $\frac{di}{dt}$ is negative. when V is negative, the opposite is true; so current is increasing and decreasing and increasing and decreasing in this range. it seems like a dampened sine wave, like what one would see in a RLC circuit. I think that this is essentially what happens; power is being dissipated as it would be in an RLC circuit but in an unideal case (ideal RLC circuit would have this sinusoidal oscillation forever, unideal/real has this dampened 'decreasing exponential' type of oscillation).

Basically, in the time after the inductor is fully discharged, the MOSFET does not switch again, which allows for RLC oscillation. The capacitor starts to discharge but some of the excess current flows back around the full circuit loop and the inductor tries to oppose this. This slightly charges the inductor, which then transfers some energy to the capacitor. The capacitor then gives more energy to the inductor, and this cycle continues all while power is being dissipated accross the resistors and likely the inductor...? Hence the lower 'boosted' voltage. I think that eventually, there will be a period long enough where all of the power would be lost, and the boost will not longer boost at all; it will be a 2V output too.

the equivalent circuit would look something like this:

In [14]: `InlineImage('cutoff.png', width=12, height=12, caption= "Figure 12. equivalent circuit while MOSFET is off in a`

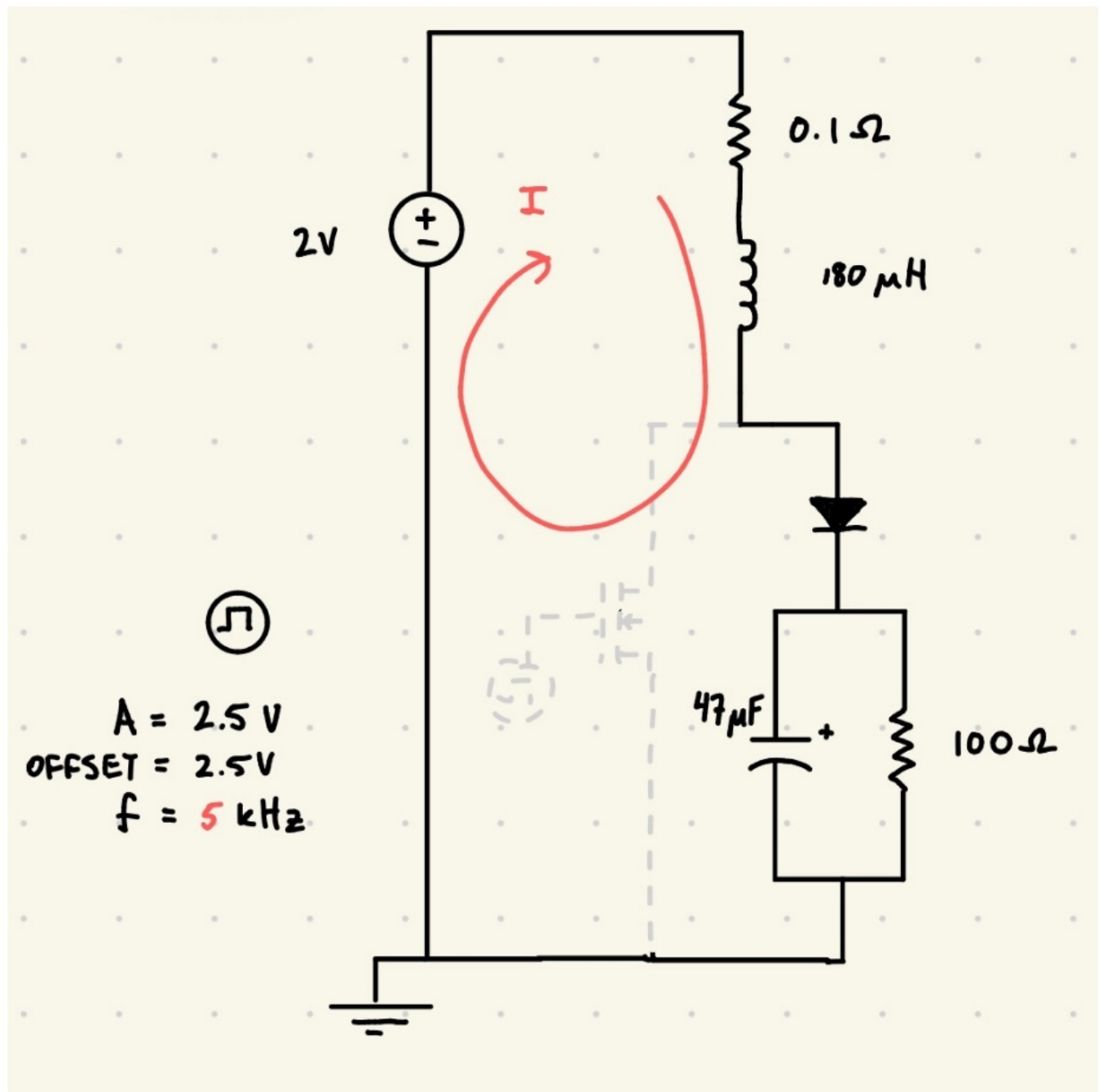


Figure 12. equivalent circuit while MOSFET is off in a 5kHz MOSFET input cycle

###2.1.3 Power Efficiency

Here, I have a few measurements.

- Average voltage over the shunt resistor = 10.089 mV
- Average voltage over the load resistor = 3.4204 V
- Range of fluctuation for the shunt resistor = 22.2146 mV
- Range of fluctuation for the load resistor = 102.214 mV

The range of fluctuation was measured using the cursors to surround the range of fluctuation for either voltage reading. I picked a relatively flat part, and zoomed in a bit more to see the zig-zag ish shape. Given that these are technically analog measurements (me moving the cursor with my mouse), I will divide by $2\sqrt{6}$ accordingly. the 2 is to divide the range of fluctuation in half.

Below are my calculations for I_{out} (current from power source), I_{R_L} (current over the load resistor), power from the 2V source, and power dissipated through the resistors:

In [5]: `InlineImage('valuecalcs.png', width=12, height=12, caption="Figure 13. Current and power calculations", fs=12)`

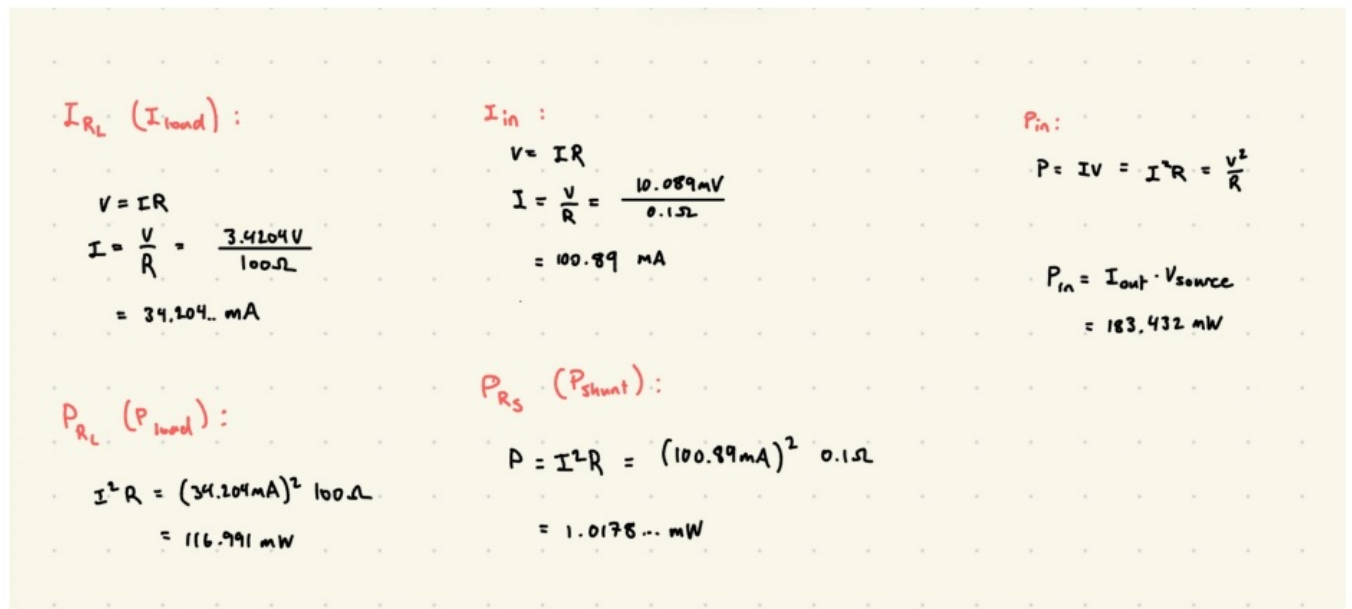


Figure 13. Current and power calculations

here is my power efficiency:

In [6]: `InlineImage('efficiencycalc.png', width=12, height=12, caption="Figure 14. power efficiency calculations", fs=`

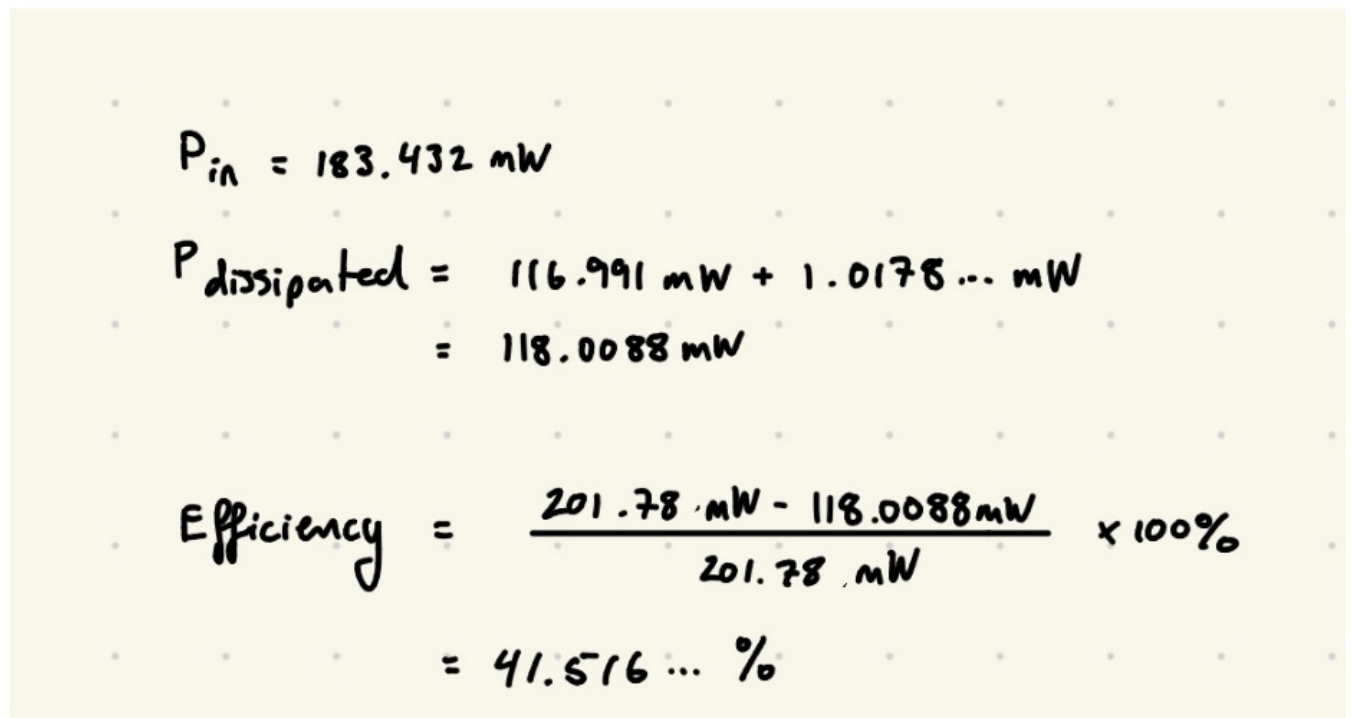


Figure 14. power efficiency calculations

And here is my uncertainty propagation: Note, some values I have not specified earlier. These include the value of the load resistor, measured at 99.2 ohms, the shunt resistor, measured at 0.092 ohms, and the voltage supply with a fluctuation of 0.006 volts. I have included uncertainty for all of these values accordingly:

In [8]: `InlineImage('uncertcalcs.png', width=12, height=12, caption="Figure 15. uncertainty propagation for power effi`

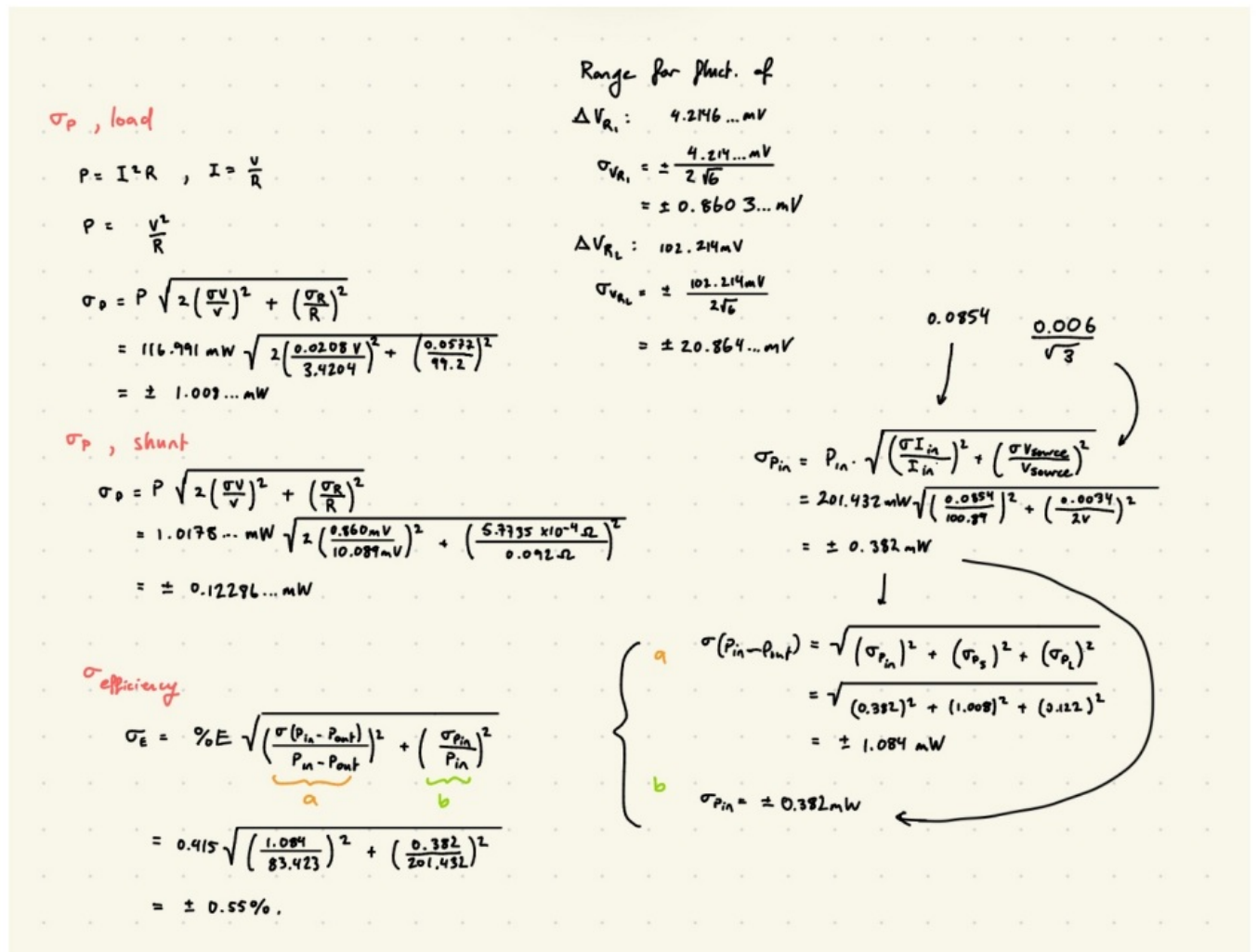


Figure 15. uncertainty propogation for power efficiency

This gives me a final power efficiency of $41.5 \pm 0.6 \%$.

###2.1.4 Load Effects

Decreasing the load resistance to 50 ohms resulted in a lower voltage boost. that is the first thing I noticed. Then, I repeated the graphs of all of the voltages and obtained the following:

In [10]: `InlineImage('newwaves.png', width=12, height=12, caption="Figure 16. Voltage waveforms for the 50 ohm dc dc boost")`

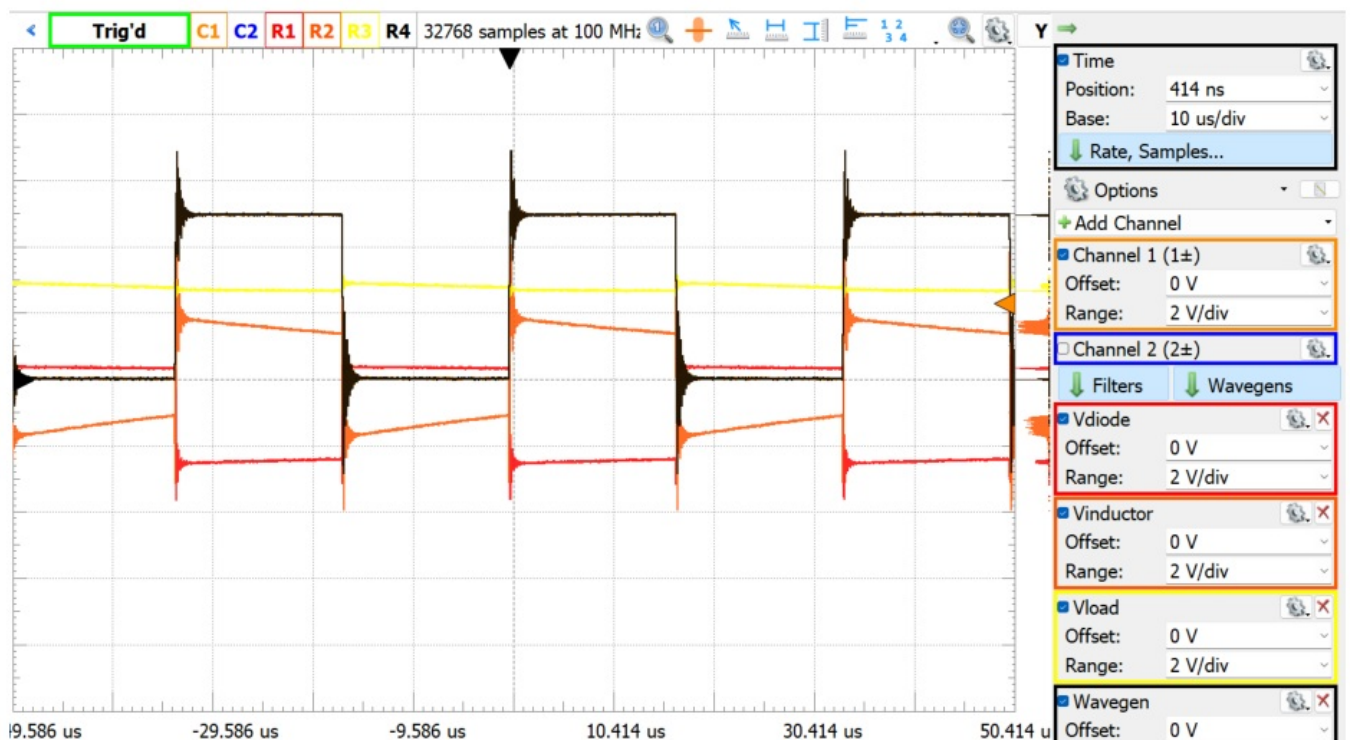


Figure 16. Voltage waveforms for the 50 ohm dc dc boost. colours and scope settings are identical as they are in the 100 ohm case for easy analysis.

Another thing I notice is that the amplitudes of all the voltages are lower. Sure, that makes sense. we still satisfy KVL. However, the major difference is that the transitions are a lot more abrupt; there is no gradual increase in slope for the voltages over the inductor and diode as there were with a $100\ \Omega$ load. At first, I am not quite sure how this happens. Going back, I think I said that my guess for the change in slope was due to the inductor running low on charge. However, I no longer think that is the case as much. In this case, I think that the inductor

When I halve the load resistance, the current needs to double for the same amount of voltage over it. However, given that the inductor is still the same, and can only output a finite quantity of energy, I think that the boost from the inductor is no longer enough to give the same boost. Hence the lower output voltage. This logically makes sense and I think the lower resistance represents a heavier load actually. With lower resistance, more current is required/drawn, which simulates a more power hungry device. So, in general, the voltage boost will boost less and less for a heavier and heavier load (assuming the other components in the boost circuit remain the same).

Hey, this leads to another thought. I think that in a dc dc boost, likely the limiting factor is the inductor and how much current it can supply. Up until a point, of course; eventually the capacitor will no longer be able to hold all of the energy given by the inductor, and the current will have to go somewhere else (or something dangerous could..?? occur??). I wonder what the limit to a dc dc boost is given the circuit components we have access to.

#3. Conclusion

In this lab, we built and looked at a dc dc boost circuit, as well as individual component behaviour (specifically the voltage fluctuations). With this, we had to reason how the circuit worked and what each component contributes.

- inductor provides the actual boost
- capacitor holds the boost provided so the boost is maintained
- diode helps keep the charge at the load We then looked at how frequency affects the circuit. As we decrease it, the inductor can charge more and provide more energy to boost the voltage. However, if we let the frequency get too low, we see that the inductor seems to discharge fully, and then begin an RLC oscillation (except the current only flows in a single direction) that takes the form of a decreasing sinusoidal waveform. Capacitor discharges over the load and the current flows into the discharged inductor, which initially opposes the current (negative voltage), then gradually lets it through, charging the inductor a bit in the process. then, the inductor discharges (positive voltage) and the capacitor charges... etc. all the while, power is being dissipated; decreasing exponential sinusoidal.

This was a good visual and real life example of what we have learned last year about circuits; actually seeing it work (in a circuit that is actually applicable to something too) solidifies the concept more.

So for the future, it is important to set the right frequency; otherwise we may end up not having a voltage boost, and we just get the input voltage again. (or potentially lower..?).

Then, we looked at the power efficiency of the voltage boost circuit. I found that this circuit is quite inefficient, and the majority of the power dissipated comes from the load resistor. Of course, this is also under ideal conditions, so we have to consider that power may be dissipated through the wires, the breadboard, the other 'non conductors' (they may not be perfect insulators), and even the mosfet potentially. I can think of a case where the MOSFET is on, and the inductor is fully charged, and so some energy may leak to GND without ever reaching the load side of the diode. The inductor also likely has some amount of internal resistance, causing unwanted energy loss to heat..?

Essentially, for the DC-DC boost to be effective, I think the number one goal for our case is to increase the efficiency; we get maybe 1.6-1.7x ish the voltage out, while losing more than half the power we put in.. Of course, in this case, the resistor is simply a representation of the load, but the operating conditions seem close enough that this power dissipation is an accurate representation of a real load that draws the same amount of current at the same voltage.

Finally, we looked at what changing the load resistance does. I noticed that the voltage drops, since the new load requires more current to achieve the same voltage out. This showed me that in our case, the inductor seems to be limiting the output voltage with how much energy it can store and then pass to the capacitor. So in the future, in designing similar devices, I think that we should choose the inductor and capacitor parts well to ensure the load can have the required voltage and current. We should also choose the load properly to be used with a given dc dc boost circuit.

To conclude, I learned a lot about how dc-dc boosts actually function in principle, and got to see what varying loads and MOSFET frequency does. I now have in my mind considerations for the future that I will be aware of should I need to build a similar circuit.

```
In [3]: # comments in Python start with a "#" and will not be executed - well commented code is easier for graders to u
# Importing modules that will be necessary for the tutorial
import numpy as np # numpy is a general math/array manipulation library that is almost always used in any scien
import matplotlib.image as mimg # matplotlib.image is used for displaying images in python
import matplotlib.pyplot as plt # matplotlib.pyplot is used for general plotting and is very similar to matlab
import pandas as pd # pandas is a data manipulation library, we will use it to import data from excel files
print('All necessary modules imported!')
```

All necessary modules imported!

```
In [4]: def InlineImage(img, width=8, height=8, caption=None, fs=24):
# Author: BG
```

```
# Renders an image with a caption underneath using matplotlib
# Requires matplotlib.pyplot to be imported as plt and matplotlib.image to be imported as mimg
# img - string, the location of an image file
# width, height - numbers, the dimensions of the figure the image is displayed in. Original aspect ratio of
# caption - string, the caption to be displayed beneath the image
# fs - number, the font size of the caption
fig,ax = plt.subplots(figsize=(width,height))
ax.imshow(mimg.imread(img))
ax.set_xlabel(caption,fontsize=fs)
ax.tick_params(
    which='both',
    bottom=False,
    top=False,
    left=False,
    labelleft=False,
    labelbottom=False)
ax.spines["top"].set_visible(False)
ax.spines["left"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["bottom"].set_visible(False)
```

In []: